

COMPUTER SCIENCE 2024

(Science General Group)

TIME: 3 Hours

SECTION 'A' (Multiple Choice Questions) (75 Marks)

NOTE: i) This section consists of 15 part questions and all are answered. Each part questions carries 1 mark.

ii) Do not copy the part questions in your answerbook. Write only answer in full against the proper number of the question and its part.

1 Choose the correct answer for each from the given options:

i) This is an object-oriented language:

* Assembly * C *

* Pascal

* C++

ii) It is called the brain of computer:

* CPU✓

* MAR

* MDR

* IR

iii) The term PUSH in STACK means to:

* Add Element✓

* Delete Element

* Update Element

* Find Element

iv) It is unauthorized duplication of computer software:

* Switching

* Networking

* Software Piracy✓

* Cyber Security

v) This virus duplicates itself:

* Trojan Horses

* Worm✓

* Time Bomb

* File Infector

vi) A telephone conversation is an example of:

* Simplex

* Half-Duplex

* Full Duplex✓

* IMpex

vii) Unix is a / an:

* Operating System✓

* Virus

* Compiler

* API

viii) It is used to develop a web page:

* HTML✓

* HTTP

* BASIC

* WWW

ix) This software displays web page:

* Compiler

* Browser✓

* Text Editor

* Spread Sheet

x) This interface allows user to interact with the computer by typing command:

* CLI✓

* GUI

* CPI

* MBR

xi) Conversion of computer code into a secret code is called:

* Normalization

* Encryption✓

* Fine Code

* Data Centralization

xii) This is the scanning input device:

* Mouse

* OMR✓

* Daisy Wheel

* Inkjet Printer

* Malware

* Software✓

* Firmware

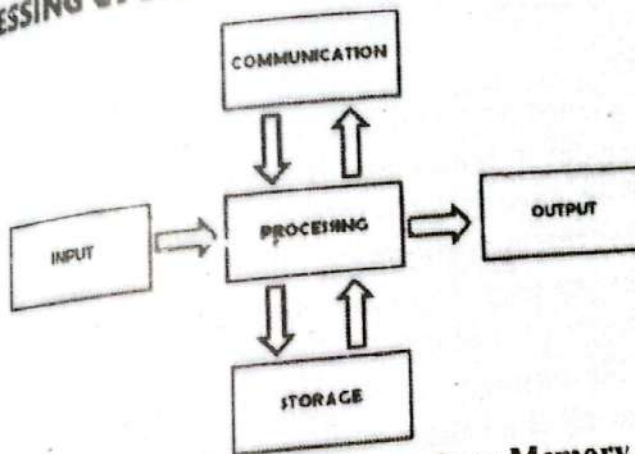
* Footwear

SECTION "B" (Short Answer Question) Marks: 30

Attempt any TEN part questions in all. All part questions equal marks. Answer should not exceed 6 to 7 lines

in a labelled diagram of Data Processing Cycle.

QUESTION: DATA PROCESSING CYCLE



Differentiate Primary Memory and Secondary Memory.

QUESTION:

PRIMARY MEMORY	SECONDARY MEMORY
Primary memory is organized as volatile & non-volatile memories.	The secondary memory is always a non-volatile memory.
It is directly accessed by the processing unit.	Data cannot be accessed directly by the processor. It is first copied from secondary memory to primary memory. Only then CPU can access it.
Volatile memory meaning data cannot be retained in case of power failure.	It's a non-volatile memory so that data can be retained even after power failure.
Primary memory can be accessed by the data bus.	Secondary memory is accessed by I/O channels.

iii) What is Cache? Mention its types.

ANSWER:

CACHE

Cache is an intermediate storage area, which is available in microprocessor. The immediate processed information is stored in cache.

TYPES OF CACHE:

The cache inside the microprocessor is called internal cache and outside called external cache.

iv) What is Language Translator? Define Interpreter.

ANSWER:

LANGUAGE TRANSLATOR :

It is another system software which convert the high level language machine level language for the purpose of machine understanding.

The Machine can only understand the machine level language or binary language 0's & 1's. The language translator rectify the errors within the program through different ways.

INTERPRETER

Interpreter is a translator which is used to convert programs in high-level language to low-level language. Interpreter translates line by line and reports the error once it encountered during the translation process. It directly executes the operations specified in the source program when the input is given by the user. It gives better error diagnostics than a compiler.

v) Why star topology is the most reliable?

ANSWER: Star topology is the most reliable because if one cable or device fails then all the others will still work. It is high-performing as no data collisions can occur. It is less expensive because each device only needs one I/O port and wishes to be connected to the hub with one link.

vi) Define the following:

- * Simplex Mode
- * Half-Duplex Mode
- * Full-Duplex Mode

SIMPLEX MODE

In Simplex mode, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit, the other can only receive. The simplex mode can use the entire capacity of the channel to send data in one direction.

HALF-DUPLEX MODE

In half-duplex mode, each station can both transmit and receive, but not at the same time. When one device is sending, the other can only receive, and vice versa. The half-duplex mode is used in cases where there is no need for communication in both direction at the same time. The entire capacity of the channel can be utilized for each direction.

FULL-DUPLEX MODE

plex mode, both stations can transmit and receive simultaneously. In plex mode, signals going in one direction share the capacity of the channel. In signals going in other direction, this sharing can occur in two

the link must contain two physically separate transmission paths, one for sending and other for receiving.

plex mode is used when communication in both directions is used all the time. The capacity of the channel, however must be divided

the two directions.

is Synchronous Transmission?

SYNCHRONOUS

ynchronous data transfer, the basic sender and receiver use the same clock signal. Hence, the sender and receiver receive the same clock signal,

to happen a special type of clock signal is required. In the

ynchronous data transfer, the master or the main sender is supposed to

the clock signal to all the receivers or recipients. The Synchronous

communication works much like a phone call. Here, the Receiver must be

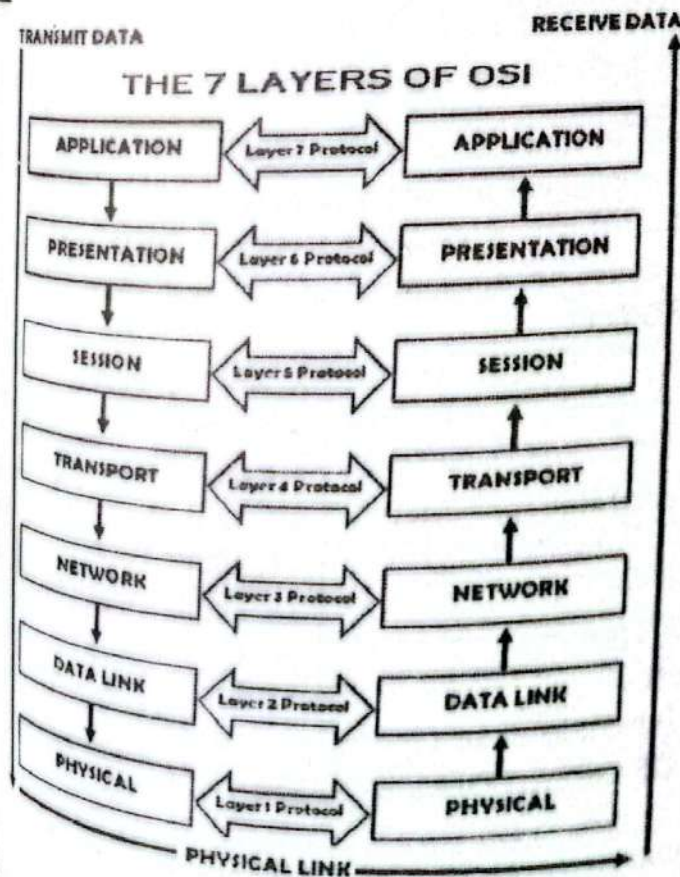
otherwise the conversation cannot occur. The synchronous

communication gives an immediate and spontaneous response from the

receiver. Hence, this type of collaboration is considered to be ideal.

is a labelled diagram of OSI reference model.

Fig:



ix) What is ASCII coding scheme?

ANSWER:

ASCII CODING SCHEME

ASCII, stands for American Standard Code for Information Interchange. It is a 7-bit character code where each individual bit represents a character. Characters in ASCII encoding include upper- and lower-case letters A through Z, numerals 0 through 9 and basic punctuation symbols. It also uses some non-printing control characters that were originally intended for use with teletype printing terminals.

x) Define the term 'BUS' with two examples.

ANSWER:

BUS

A bus is a common pathway through which information flows from one computer component to another. This pathway is used for communication purpose and it is established between two or more computer components.

EXAMPLES OF BUS

CONTROL BUS

Control Bus carries command between different components to control activities in a computer.

DATA BUS

Data Bus carries data between the processor, memory unit and other components.

xi) What is a Port? Enlist its types.

ANSWER:

PORT

A computer port or port is a connection point or interface between a computer and an external or internal device. Internal ports may connect such devices as hard drives and CD ROM or DVD drives; external ports may connect modems, printers, mice and other devices.

TYPES OF PORT

- | | |
|---------------------------------------|--|
| 1. Serial Port | 8. Modem Port |
| 2. Parallel Port | 9. Connects a PC's modem to the telephone network. |
| 3. PS/2 Port | 10. Ethernet Port |
| 4. Universal Serial Bus (or USB) Port | 11. Game Port |
| 5. VGA Port | 12. Digital Video Interface, DVI port |
| 6. Power Connector | 13. Sockets |
| 7. Firewire Port | |

xii) Enlist five functions of an Operating System.

FUNCTIONS OF OPERATING SYSTEM

ANSWER:

Operating system performs the following functions:

ing and Execution
Management
Controlling
interface
PUSH and POP in STACK.

2. Memory Management
4. Data security
6. Process Management
8. Printing controlling

Q:

operation is used to insert an item in the stack. Push operation also checks whether the stack is empty and if it is empty then Push operation inserts an item into the stack. Once the stack is full and there is no more space to add an item, the push operation will inform that the stack is in overflow state. So, no new item can be inserted into the stack.

operation is used to delete or remove an item from the stack. Pop operation also performs whether the stack is full or not. If the stack is full, pop operation will delete the item one by one from the stack till the stack is empty. Once the stack is empty, Pop operation will return the error as the stack is in underflow state. So, there is no item in the stack to delete or remove.

Stack is a concept mainly used in data structures. Stack is used with the LIFO concept and used for data inserting, modifying and recovering. Stack uses the concept of LIFO (Last-In-First-Out). LIFO refers, stack operations will be performed on the data that is in the last part of the array. Stack can also be used in linked-list but at the low level compared with array.

What is an Email? Write three advantages of Email.

Q:

Email is defined as the transmission of messages over communications networks. Typically the messages are notes entered from the keyboard or saved as files stored on disk. Most mainframes, minicomputers, and local area networks have an email system.

Electronic mail systems are confined to a single computer system or network, but others have gateways to other computer systems, enabling them to send electronic mail anywhere in the world. Companies that are computerized make extensive use of e-mail because it is fast, flexible, and reliable.

Write the full form of any six of the following abbreviations:

- | | | |
|-------|-------|----------|
| * LCD | * MBR | * IR |
| * ALU | * CPU | * TCP/IP |

ANSWER:

ABBREVIATION	FULL FORM
* LED	Light-Emitting Diode
* LCD	liquid-crystal display
* MBR	Master Boot Record
* IR	Instruction register
* BIOS	Basic Input Output
* ALU	Arithmetic and logic operations
* CPU	Central Processing Unit
* TCP	Transmission Control Protocol
* IP	Internet Protocols

Section C (Detailed –Answer Question) (30)

NOTE: Attempt any THREE question from this section.

All questions carry equal marks.

3. What is a Monitor? Explain CRT and Flat Panel Monitors with the types.

ANSWER:

MONITORS

The most common computer output device is the monitor or computer screen. Monitors create a visual display for you to view from processed data. They come in a variety of screen sizes and visual resolutions. There are two common types of modern computer monitors, cathode ray tube and flat panel screens.

CRT monitors use phosphorescent dots to create the pixels that make up displayed images. Flat panel monitors usually use liquid crystals or plasma to create output. Light is passed through liquid crystals to create the pixels. All monitors rely on a video card that is located on the computer motherboard or in a special expansion slot. The video card processes the computer data into image details that the monitors can display.

TYPES OF MONITORS

CATHODE RAY TUBE (CRT) MONITORS

It is a technology used in early monitors. It uses a beam of electrons to create an image on the screen. It comprises the guns that fire a beam of electrons inside the screen. The electron beams repeatedly hit the surface of the screen. These guns are responsible for generating RGB (Red, Green, Blue) colors, and more other colors can be generated with the help of combining these three colors. Today's Flat Panel Monitors replace the CRT monitors.

FLAT PANEL MONITORS

These types of monitors are lightweight and take less space. They consume less power as compared to CRT monitors. These monitors are more

as they do not provide harmful radiation. These monitors are expensive than CRTs.

CRYSTAL DISPLAY:

Liquid crystal display) screen contains a substance known as liquid crystal. The particles of this substance are aligned in a way that the light passing through the backside on the screens, which allow to generate an image or block. Liquid crystal display offers a clear picture as compared to CRT display and emits less radiation. Furthermore, it consumes less power and takes less space than a CRT display.

PLASMA DISPLAY:

Plasma display uses gas plasma technology, which uses a layer of gas between two layers of glass. When voltage is applied, the gas releases ultraviolet light. This ultraviolet light, the pixels on the screen glow and form an image. Plasma displays are available in different sizes of up to 150 inches. Although they show effective colors as compared to the LCD monitor, it is more expensive. That's why it is less used.

different types of Printers.

TER:

OF PRINTERS

are divided into two major types, impact and non-impact printers. Impact printers are further divided into different types:

1. Impact printer

Impact printer has mechanisms resembling those of a typewriter. It prints characters or images by striking a mechanism such as a print head or wheel against an inked ribbon, leaving an image on paper.

OF IMPACT PRINTERS

are some types of impact printers.

MATRIX PRINTERS

Dot-matrix printer uses print heads containing from 9 to 24 pins. These pins produce patterns of dots on the paper to form the individual characters. The 24 pin dot-matrix printer produces more dots than a 9 pin dot-matrix printer, which results in much better quality and clearer characters. The general rule is: the more pins, the clearer the letters on the paper. The pins strike the ribbon individually as the print mechanism moves across the entire print line in both directions, i.e., from left to right, and so on. The user can produce a color output with a dot-matrix printer (the user will change the black ribbon with a ribbon of different colors). Dot-matrix printers are inexpensive and typically print speeds of 100-600 characters per second.

DAISY-WHEEL PRINTERS

In order to get the quality of type found on typewriters, a daisy-wheel impact printer can be used. It is called daisy-wheel printer because its print mechanism looks like a daisy; at the end of each "Petal" is a formed character which produces solid-line print. A hammer strikes a "petal" containing a character against the ribbon, and the character prints on the paper. Its speed is slow typically 25-55 characters per second.

LINE PRINTERS

In business where enormous amount of material are printed, the character-at-a-time printers are too slow; therefore, these users need line-at-a-time printers. Line printers, or line-at-a-time printers, use special mechanism that can print a whole line at once; they can typically print the range of 1,200 to 6,000 lines per minute. Drum, chain, and band printers are line-at-a-time printers.

DRUM PRINTER

A drum printer consists of a solid, cylindrical drum that has raised characters in bands on its surface. The number of print positions across the drum equals the number available on the page. This number typically ranges from 80-132 print positions. The drum rotates at a rapid speed. For each possible print position there is a print hammer located behind the paper. These hammers strike the paper, along the ink ribbon, against the proper character on the drum as it passes. One revolution of the drum is required to print each line. This means that all characters on the line are not printed at exactly the same time, but the time required to print the entire line is fast enough to call them line printers. Typical speeds of drum printers are in the range of 300 to 2000 lines per minute.

CHAIN PRINTERS

A chain printer uses a chain of print characters wrapped around two pulleys. Like the drum printer, there is one hammer for each print position. Circuitry inside the printer detects when the correct character appears at the desired print location on the page. The hammer then strikes the page, pressing the paper against a ribbon and the character located at the desired print position. An impression of the character is left on the page. The chain keeps rotating until all the required print positions on the line have filled. Then the page moves up to print the next line. Speeds of chain printers range from 400 to 2500 characters per minute.

BAND PRINTERS

A band printer operates similar to chain printer except it uses a band instead of a chain and has fewer hammers. Band printer has a band divided into five sections of 48 characters each. The hammers on a band printer are mounted on a cartridge that moves across the paper to the

Characters are rotated into place and struck by the foot styles can easily be changed by replacing a band or chain.

IMPACT PRINTERS

Impact printers, used almost everywhere now, are faster and quieter than non-impact printers because they have fewer moving parts. Non-impact printers form characters and images without direct physical contact with the printing mechanism and the paper.

NON-IMPACT PRINTERS

Types of non-impact printers are following:

LASER PRINTER

Laser printer is a non-impact printer. Its working is similar to a photocopier. It uses laser beams to burn special powder on page to produce permanent image. The power is contained in toner.

INKJET PRINTER

Inkjet printer is a type of non-impact printer. It prints characters and graphics by spraying tiny drops of liquid ink on paper. These printers can produce high quality text and graphics in both black-and-white and color. A typical inkjet printer provides resolution of 300 dots per inch. The latest inkjet printers provide higher resolution.

DYE-SUB PRINTER

Dye-sub printer uses color waxes and heat to produce images on special color wax sheets. These sheets are not required for black-and-white output. These printers are expensive and require costly paper. These printers create the highest-quality color printing.

Software Privacy? How can we protect our Software?

SOFTWARE PRIVACY

Software piracy is the unauthorized use of legally protected software. It includes stealing, copying, distributing, modifying or selling the software. Copyright laws were originally put into place so that people who create software (programmers, writers, graphic artists, etc.) would get fair compensation for their work. When software piracy occurs, compensation is stolen from these copyright holders.

WAYS TO PROTECT:

There are many ways to protect yourself from unintentionally pirating software or infecting your computer with malware, including:

1. YOUR LICENSE KEYS:

Always have the correct license keys and follow the proper protocols when installing your software.

Purchasing software from authorized retailers: Buy software only from credible vendors and stores; avoid third-party or noncredible sources to ensure you receive legitimate products.

USING THE LATEST WI-FI SECURITY PROTOCOLS:

Ensure your router uses the latest software to prevent cybercriminals from breaching your data or infecting your computer with viruses.

REPORTING SUSPICIOUS SOFTWARE:

If you encounter strange or suspicious software you believe may be malicious, report your findings to The Software Alliance (BSA).

6. What are Scanning Devices? Explain their different types.

SCANNING DEVICES

Scanning devices use light-sensitive equipment to record data in the form of bar codes, optical marks, typewritten characters, or magnetic symbols, or even just plain text or graphics. This data is then converted into digital format.

SCANNING DEVICES

Following devices are used for scanning purposes.

SCANNER

A scanner is a device that captures images from photographic prints, posters, magazine pages, and similar sources for computer editing and display. Scanners come in hand-held, feed-in, and flatbed types and scan in black-and-white only, or color. Very high resolution scanners are used for scanning for high-resolution printing, but lower resolution scanners are adequate for capturing images for computer display. Scanners usually come with software, such as Adobe's Photoshop program, that lets you resize and otherwise modify a captured image.

BAR-CODE READER:

Bar codes are the vertical zebra-stripped marks we find on most of the manufactured products in the market. This bar-code system is also called the "Universal Product Code". These are read by bar-code readers, which are photoelectric scanners that translate the bar-code symbols into digital code, which is then fed to the computers for further processing.

FAX MACHINE:

The fax or facsimile transmission machine scans an image and sends it as electrical signals over telephone lines to a receiving fax machine, which then creates the image on paper. We have two types of fax machines: dedicated fax machines and fax modems. Dedicated fax machines are normal fax machines whereas, the fax modem is a circuit board inside a computer system unit. It has a capability to send signals directly to someone else's machine or computer fax modem, from computer to computer.

7. Define Computer Virus. Explain its types.

ANSWER:

COMPUTER
PAPERS XI (2)
COMPUTER VIRUS
A computer virus is a program, script, or macro designed to cause damage, steal personal information, modify data, send e-mail, display messages, or perform a combination of these actions.

KINDS OF COMPUTER VIRUS

PERMANENT VIRUSES:

Permanent viruses dwelling in RAM memory. In this case, they are in a position to overcome, as well as interrupt, all operations that the computer executes. Their effects include corrupting programs and files, deleting, opening, renaming or copying.

DELETE VIRUSES:

These delete information that is in the infected files. In this case, the files would be rendered totally or partially useless. Unfortunately, they can only clean the infected file by deleting it completely, therefore the original content is lost.

REPLICATION VIRUSES:

These replicate themselves, then act when executed. Subject to satisfaction of particular conditions, the virus infects files located in the folders or sub-directories.

TRIGGER VIRUSES:

These infect executable files or programs. On running the programs, the virus would be activated, then be able to carry out its damaging effects. The existing viruses are in this category.

BOOT VIRUSES:

These infect the hard disk's or floppy drive's boot sector. This would render the computer unable to boot. These viruses can, however, be avoided by ensuring that the floppy disks and hard drive is well protected. Never use a computer using an unknown disk drive or floppy disk.

PATH VIRUSES:

These alter the paths indicating a file's location. In this case, when the program is executed, we will be running the program from a different location, since the virus has moved the original program and file to a new location.

MACRO VIRUS:

This affects files created using particular programs or applications such as word processing macros. The mini-programs increase their ability to automate actions, in which case they would be performed as single actions.

WORM VIRUS:

These are even if most are Macro viruses. Generally these kinds of viruses spread through social interaction with someone who e-mails the user.

RANSOMWARE:

Ransomware is type of malicious software product that attempts to lock computer for ransom. This might be locking files, hiding files, or denying permissions on files or encrypting files.

COMPUTER SCIENCE 202

TIME: 3 Hours

SECTION 'A' (Multiple Choice Questions)

(75 Marks)

1 Choose the correct answer for each from the given options

i) This is an example of system software:

- * MS Excel * MS Word * Operating System ✓ * Basic

ii) This is a low level language:

- * Assembly ✓ * Visual Basic * Java * C++

iii) This is a sequential storage medium

- * CD * Magnetic Tape ✓
* Punch Card * Hard disk

iv) This topology has central hub:

- * Mesh * Star ✓ * Ring * Bus

v) The process of converting digital signals to analog signals is called

- * Modulation ✓ * Demodulation
* Synchronization * Transformation

vi) This is a set of rules that govern data communications:

- * Buses * Registers * Routers * Protocols ✓

vii) This is not a computer virus:

- * Trojan Horse * Time Bomb
* Avast ✓ * Worm

viii) This is a computer virus that replicates itself:

- * Bug * Worm ✓ * Vaccine * Bomb

ix) Bluetooth is an example of:

- * Personal area Network ✓ * Local Area Network
* Virtual Private Network * Wide Area Network

x) This is a printer that works by spraying ionized ink at sheet of paper

- * Ink-jet Printer ✓ * Dot matrix
* Laser Printer * Chair Printer

xi) In topology all of the nodes in the network are connected to a single cable or line:

- * Ring * Star * Tree * Mesh

Correct answer is Bus

xii) A television broadcast is an example of this transmission mode:

- * Half Duplex * Simplex ✓

- * Multiplex
- * Decimal Per Inch
- * Data Per Inch

A group of bits that tells the computer to perform a specific operation is known as:

- * Algorithm
- * Register.
- * Pirate
- * Thief

A person who gains illegal access to a computer system is known as:

* Spy

SECTION "B" (Short Answer Question) Marks: 30

Attempt any TEN part questions in all. All part questions are of equal marks. Answer should not exceed 6 to 7 lines

What is data communication? Name its components

ANSWER:
DATA COMMUNICATION refers to the transmission of this digital data between two or more computers and a computer network or data network. A telecommunications network that allows computers to exchange data. A physical connection between networked computing devices is established using either cable media or wireless media. The best-known computer network is the Internet.

COMPONENTS OF DATA COMMUNICATION:

There are mainly five components of a data communication system:

1. Message
2. Sender
3. Receiver
4. Transmission Medium
5. Set of rules (Protocol)

Write any three differences between Impact Printer and Non-Impact printer.

ANSWER:

DIFFERENCES BETWEEN IMPACT PRINTER AND NON-IMPACT PRINTER

IMPACT PRINTERS	NON-IMPACT PRINTERS
The print quality of an impact printer is lower.	The non-impact printers have a higher print quality.
Impact printers use continuous sheets of paper.	Non-impact printers use individual sheets of paper.
Impact printers depend upon traditional printing methods.	Non-impact printers depend upon the most modern printing techniques.

What is a Computer Virus? How can we protect our computer from virus?

ANSWER:

VIRUS

A computer virus is a program, script, or macro designed to cause damage, steal personal information, modify data, send e-mail, display messages, or some combination of these actions.

PROTECTION OF COMPUTER:

The only solution to never risk getting a computer virus is to keep your computer disconnected from the internet and off which is not use. We have many options that can greatly reduce the risk of getting a computer virus.

1. Keep our operating system up to date. Microsoft, Apple and Linux have updates that roll out on a regular basis. Keeping our operating system up to date alone will stop the vast majority of known threats.
 2. Keep our web browser updated. Nearly all web browsers have an auto update feature that allows us to auto update our web browser. Popular web browsers out there include, FireFox, Chrome, Safari and Edge (formerly known as Internet Explorer)
 3. Keep our software programs updated. Programs such as Office, Adobe PDF, Flash are all targeted programs due to their wide user base. The more popular a program is the more it's a target for exploits.
- iv) What is meant by Modulation and De-modulation?

ANSWER:**MODULATION**

Modulation is a process through which audio, video, image or text information is added to an electrical or optical carrier signal to be transmitted over a telecommunication or electronic medium. Modulation enables the transfer of information on an electrical signal to a receiving device that demodulates the signal to extract the blended information.

DEMODULATION

When digital information is transmitted over an analog medium, such as a telephone line, an electronic waveform is modulated to contain the information. This analog signal is referred to as a carrier wave. Demodulation is the act of extracting digital data from this signal once it reaches its destination. A common example of a demodulating device is a modem.

v) Which technical issue is common in Ring and Bus Topology?

ANSWER: Bus and ring both topologies have the common technical issue of device failure and slower. If one device fails or gets disconnected, the whole network stops functioning, and if the number of devices grows, the network becomes slower and more congested.

vi) Why ROM is called non-volatile memory?

ANSWER: ROM or Read Only Memory refers mainly to the part of the system program that is inbuilt and required for the operating system to

properly. ROM is non volatile memory because all the data in it is not erased after shutting down the computer and restarting it.

Q. What is the function of a Compiler and an interpreter?

Ans:

COMPILER

Compiler is a translator which is used to convert programs in high-level language to low-level language. It translates the entire program and also, it checks for the errors in source program encountered during the translation.

INTERPRETER

Interpreter is a translator which is used to convert programs in high-level language to low-level language. Interpreter translates line by line and checks for the error once it encountered during the translation process. It executes the operations specified in the source program when the user gives the program. It gives better error diagnostics than a compiler.

Q. What do you mean by Broadcasting?

Ans:

BROADCASTING

Broadcasting in computer networks is a type of communication mechanism where the message to be received by all the nodes of a network. The term broadcast in general refers to the transmission of signals from radio stations.

A broadcasted signal is stopped at layer-3 network layer of OSI or to be practical - at the router. A more technical example of broadcasting would be: The Address-Resolution-Protocol request (ARP) whenever a host needs to resolve an IP address to its corresponding MAC address it will broadcast a signal asking "Who this IP belongs to?" and this broadcasted signal is received by every single node in a network domain and then an appropriate node will respond accordingly.

Q. What is the main difference between Router and Gateway?

Ans:

DIFFERENCE BETWEEN ROUTER AND GATEWAY

ROUTER	GATEWAY
Hardware device which is capable for receiving, storing and forwarding the packets to other networks.	It is a device that is used for the communication among the networks which have a different set of protocols.
The main function of a router is to direct the traffic from one network to the other.	The main function of a gateway is to translate one protocol to the other.
It is hosted on only the dedicated	It is hosted on dedicated

applications.

applications, physical servers
virtual applications.

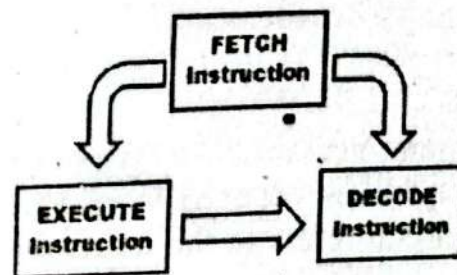
x) Name any two simplex devices used in communication system.
ANSWER: Television and radio are two simplex devices used in communication system.

xi) Why Address Bus is called unidirectional?
ANSWER: Data bus is bidirectional because it transfers information between peripheral devices and memory register but, address bus is unidirectional because it simply receives information.

xii) Draw the block diagram of Fetch Cycle.

ANSWER:

BLOCK DIAGRAM OF FETCH CYCLE



xiii) Name the fastest communication media.

ANSWER:

FASTEST COMMUNICATION MEDIA

Fibre optics cable is the fastest medium of communication. It uses the concept of refraction of light through a core made up of glass or plastic. The core is surrounded by a less dense glass or plastic covering called cladding. It is used for the transmission of large volumes of data.

xiv) Define Protocol. Name any three protocols.

ANSWER:

PROTOCOL

A protocol is a set of rules and guidelines for communicating data. Rules are defined for each step and process during communication between two or more computers. Networks have to follow these rules to successfully transmit data.

DIFFERENT PROTOCOLS

1. Transmission Control Protocol (TCP)
2. Hypertext Transfer Protocol (HTTP)
3. File Transfer Protocol (FTP)

xv) What is Copyright Law?

ANSWER:

COPYRIGHT LAW

Copyright is a legal term describing ownership of control of the rights in the use and distribution of certain works of creative expression, including

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video, movies, music and computer programs. Historically,
copyright law has been enacted to balance the desire of cultures to use and
creative works (thus creating "derivative work") against the desire
creators of art, literature, music and the like to monetize their work
controlling who can make and sell copies of the work.

Section C (Detailed - Answer Question) (30)

NOTE: Attempt any THREE question from this section.
questions carry equal marks.
What is System Software? Explain operating system with its function.

ANSWER: **SYSTEM SOFTWARE**

System software is software on a computer that is designed to control and
with computer hardware. The two main types of system software are
operating system and the software installed with the operating system,
called utility software. The operating system and utility software
depend on each other to function properly.

OPERATING SYSTEM SOFTWARE (O.S) :

Operating System is a program designed to run other programs on a
computer. A computer's operating system is its most important program.
considered as the backbone of a computer managing both software
hardware device. Operating systems are responsible for each and
everything from the control and allocation of memory to input from
external devices and output to computer display.

FUNCTIONS OF OPERATING SYSTEM

Operating system performs the following functions:

1. Booting
2. Memory Management
3. Loading and Execution
4. Data security
5. File Management
6. Process Management
7. Device Controlling
8. Printing controlling
9. Providing interface

What is Computer Network? Explain different types of network
technologies.

ANSWER:

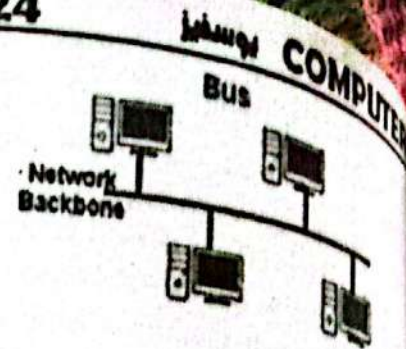
COMPUTER NETWORK

Computer network is a group of computers and related equipment
connected by a communication links to share data and other resources.
related equipment may be printer, scanners, fax machines, server, etc.
resources may include a file server, internet connection, etc.

TYPES OF TOPOLOGIES:

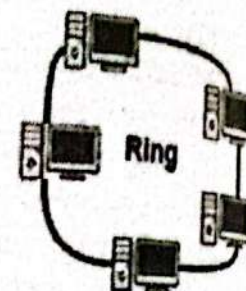
1. BUS TOPOLOGY

The physical bus topology is the simplest and most widely used of the network designs. It consists of one continuous length of cabling (trunk) and a terminating resistor (terminator) at each end. The data communications message travels along the bus in both directions until it is picked up by a workstation or server NIC.



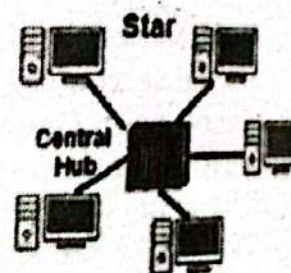
RING TOPOLOGY

As its name implies, the physical ring topology is a circular loop of point-to-point links. Each device connects directly or indirectly to the ring through an interface device or drop cable. Messages travel around the ring from node to node in very organized manner. Each workstation checks the messages for a matching destination address.



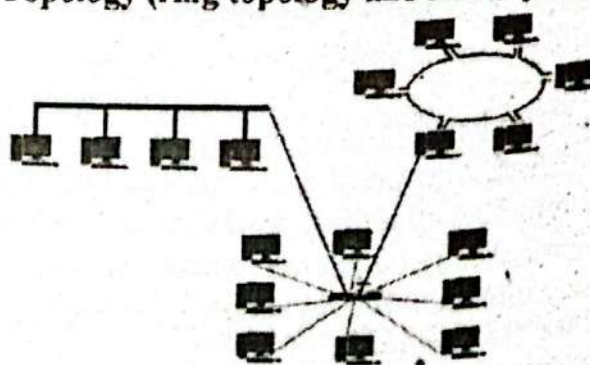
STAR TOPOLOGY

The Physical star topology uses a central controlling hub with dedicated legs pointing in all directions – like points of a star. Each network devices has a dedicated point-to-point link to the central hub. This strategy prevents troublesome collisions and keeps the line of communication open and free of traffic.



HYBRID TOPOLOGY

It is two different types of topologies which is a mixture of two or more topologies. For example if in an office in one department ring topology is used and in another star topology is used, connecting these topologies will result in Hybrid Topology (ring topology and star topology).



Explain OSI model with layered diagram.

ANSWER: OSI MODEL

The Open System Interconnection (OSI) model defines a networking framework to implement protocols in seven layers. Use this handy guide to compare the different layers of the OSI model and understand how they interact with each other.

In the OSI model, control is passed from one layer to the next, starting at the application layer in one station, and proceeding to the bottom layer, then the channel to the next station and back up the hierarchy. The OSI model takes the task of inter-networking and divides that up into what is referred to as a vertical stack that consists of the following 7 layers. Click a link in the list below to read detailed information and examples of each layer:

Physical (Layer 1)

Network (Layer 3)

Session (Layer 5)

Application (Layer 7)

2. Data Link (Layer 2)

4. Transport (Layer 4)

6. Presentation (Layer 6)

PHYSICAL (LAYER 1)

Model, Layer 1 conveys the bit stream - electrical impulse, light or radio signal — through the network at the electrical and mechanical level. It provides the hardware means of sending and receiving data on a carrier, including defining cables, cards and physical aspects. Fast Ethernet, IEEE 802.3, and ATM are protocols with physical layer components.

DATALINK (LAYER 2)

OSI Model, Layer 2, data packets are encoded and decoded into bits. It defines transmission protocol knowledge and management and handles flow control in the physical layer, flow control and frame synchronization. The data link layer is divided into two sub layers: The Media Access Control (MAC) layer and the Logical Link Control (LLC) layer. The MAC sub layer controls how a computer on the network gains access to the data and transmits it. The LLC layer controls frame synchronization, flow control and error checking.

NETWORK (LAYER 3)

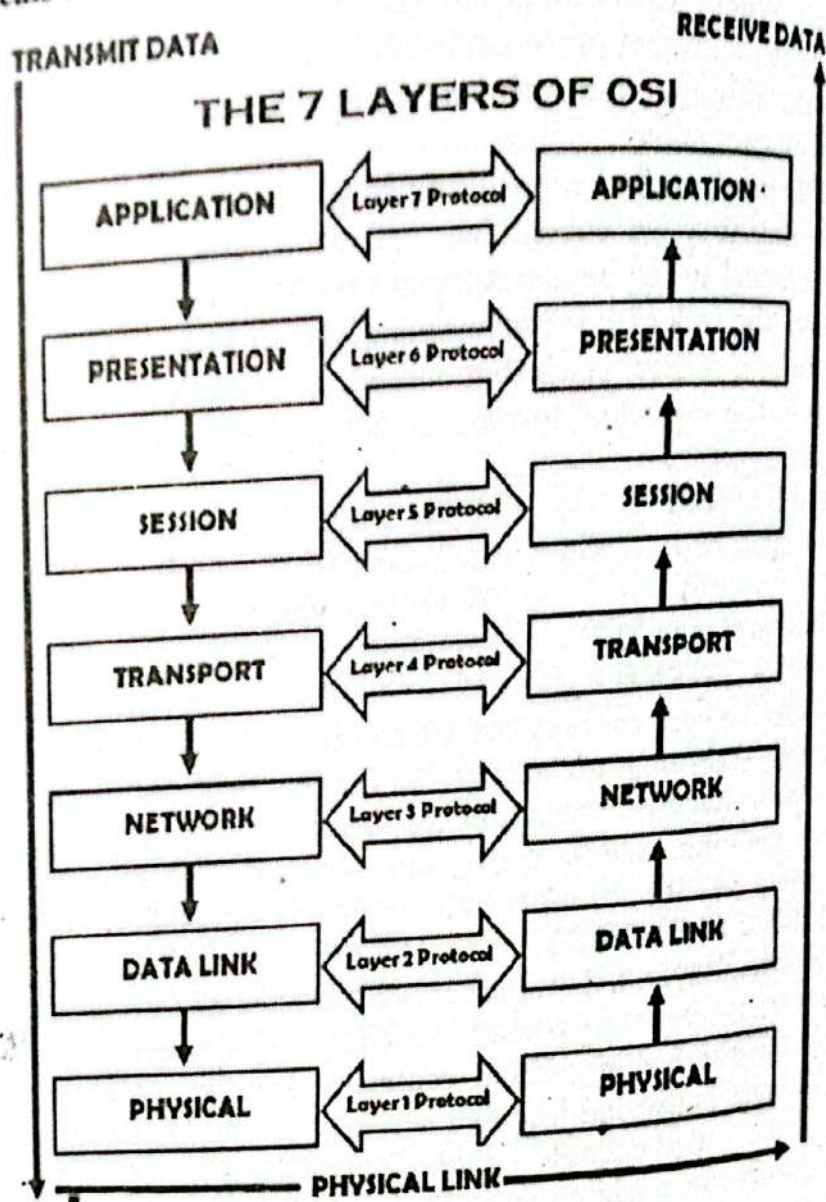
Layer 3 provides switching and routing technologies, creating logical paths, such as virtual circuits, for transmitting data from node to node. Routing and forwarding are functions of this layer, as well as addressing, error handling, congestion control and packet sequencing.

TRANSPORT (LAYER 4)

Model, Layer 4, provides transparent transfer of data between end systems or hosts, and is responsible for end-to-end error recovery and flow control. It ensures complete and reliable data transfer.

SESSION (LAYER 5)

This layer establishes, manages and terminates connections between applications. The session layer sets up, coordinates, and terminates conversations, exchanges, and dialogues between the applications at each end. It deals with session and connection coordination.



PRESENTATION (LAYER 6)

This layer provides independence from differences in data representation (e.g., encryption) by translating from application to network format, and vice versa. The presentation layer works to transform data into the form that the application layer can accept. This layer formats and encrypts data to be sent across a network, providing freedom from compatibility problems. It is sometimes called the syntax layer.

APPLICATION (LAYER 7)

OSI Model, Layer 7, supports application and end-user processes. Communication partners are identified, quality of service is identified, user

ation and privacy are considered, and any constraints on data are identified. Everything at this layer is application-specific. This provides application services for file transfers, e-mail, and other software services. Telnet and FTP are applications that exist in the application level. Tiered application architectures are part of the application level.

per. CPU Register? Explain its different types.

REGISTERS
A processor register OR CPU register) is one of a small set of data places that are part of the computer processor. Register are used to accept, store, and transfer data and instructions that are being used by the CPU, there are various types of Registers those for various purpose.

Registers are the top of the memory hierarchy, and are the fastest in the system to manipulate data. In a very simple microprocessor, it is a single memory location, usually called an accumulator. They are built from fast multi-ported memory cell. They must be able to move data onto an internal bus in a single clock cycle. The result of an operation is stored here and could be re-used in a subsequent operation or saved into memory.

VARIOUS TYPES OF REGISTERS

MEMORY ADDRESS REGISTER (MAR):

It holds the address of memory where CPU wants to read or write. When CPU wants to store some data in the memory or reads data from the memory, it places the address of the required memory in the MAR.

MEMORY BUFFER REGISTER (MBR):

It holds the contents of data or instruction read from, or written to, memory. The contents of instruction placed in this register are sent to the Instruction Register, while the contents of data are sent to the accumulator or I/O register.

ADDRESS REGISTER (I/O AR):

This register is used to specify the address of a particular I/O device.

DATA BUFFER REGISTER (I/O I3R):

This Register is used for exchanging data between the I/O module and the processor.

PROGRAM COUNTER (PC)

The Counter register is also known as Instruction Pointer Register. It is used to store the address of the next instruction to be executed. When the instruction is fetched, the value of IP is incremented.

incremented. Thus this register always points or holds the address of next instruction to be fetched.

6. INSTRUCTION REGISTER (IR):

Once an instruction is fetched from main memory, it is stored in the Instruction Register. The control unit takes instruction from this register, decodes and executes it by sending signals to the appropriate component of the computer to carry out the task.

7. ACCUMULATOR REGISTER:

The accumulator register is located inside the ALU. It is used during arithmetic & logical operations of ALU. The control unit stores data values fetched from main memory in the accumulator for arithmetic or logical operation. This register holds the initial data to be operated upon, the intermediate results, and the final result of operation. The final result is transferred to main memory through MBR.

8. STACK CONTROL REGISTER:

A stack represents a set of memory blocks; the data is stored in and retrieved from these blocks in an order, i.e. First In and Last Out (FILO). The Stack Control Register is used to manage the stacks in memory. The size of this register is 2 or 4 bytes.

9. FLAG REGISTER:

The Flag register is used to indicate occurrence of a certain condition during an operation of the CPU. It is a special purpose register with size one byte or two bytes. Each bit of the flag register constitutes a flag (or alarm), such that the bit value indicates if a specified condition was encountered while executing an instruction.

7. What is Communication Media? Discuss its different types.

ANSWER:

COMMUNICATION MEDIA

Communication media refers to the means of delivering and receiving data or information. In telecommunication, these means are transmission and storage tools or channels for data storage and transmission.

The communication media acts as a channel for linking various computing devices so that they may interact with each other. Contemporary communication media facilitate communication and data exchange among a large number of individuals across long distances via email, teleconferencing, internet forums and many other forms of communication.

TYPES OF COMMUNICATION MEDIA

1. TWISTED PAIR

It consists of a pair of copper wires twisted around each other; the wires are around 1 to 2 mm thick and they are twisted to reduce the interference from the surrounding wires. Remember that a current carrying wire has a

field around it that can interfere with that of another wire when in proximity.
by Alexander Graham Bell, this form of wire has been in use since the 19th century and is currently the cheapest medium available. Pairs are also used in telephone wires, but they only consist of four pairs. In computer networks, eight wires or four pairs are used. This is also known as the Ethernet cable or RJ-45 cable. The pairs are bundled together and covered by a protective shield.

COAXIAL CABLE
For better data rates and less signal attenuation, a coaxial cable consists of a central copper conductor that is surrounded by a foil shield. This is covered by yet another shield known as a braided shield. Unlike twisted pairs, coaxial cables only have a single copper conductor. The conductor and the foil shield are separated by a dielectric.

FIBER OPTIC
The most expensive of wired mediums and offers the highest rates of data transmission. They are often used in long distance communications and are never affected by any electromagnetic fields. This is because it transmits light.

A fiber optic cable is a thin, flexible, transparent medium made of very fine plastic fibers. It utilizes the principle of total internal reflection. Unlike twisted pairs or coaxial cables, a fiber optic uses light pulses generated by laser or an injection diode to transmit data. Each pulse represents a single bit of data.

COMMUNICATION SATELLITES
Satellite transmission is similar to microwave transmission except instead of involving another nearby microwave dish antenna, it involves a satellite many miles up in space. Figure 3.13 depicts a geosynchronous satellite. Geosynchronous means that the satellite remains stationary over one point on the earth. One disadvantage of satellite communication is the propagation delay that occurs because the signal has to travel into space and back to earth, a distance of many miles that even the speed of light can be noticeable.

COMPUTER 2022

1 Hour

(75 Marks)

SECTION 'A' (Multiple Choice Questions) (29)

This section contains of 29 part questions and all are to be attempted. Each question carries one mark.

Section "A" of Multiple Choice Questions (MCQs) must be attempted on the provided OMR sheet.

Correct answer bubble must be filled on OMR sheet.

- (iv) Use only blue/black ball point pen on OMR sheet.
 (v) Avoid pencil / White-o pen / correction pen / ink remover and gel pen to solve the questions on OMR sheet.

1 Choose the correct answer for each from the given options:

- (i) One Gigabyte (GB) is equivalent to:
 * 1024 KB
 * 1024 MB ✓
 * 1024 TB
 * 1024 ZB
- (ii) EBCDIC is the abbreviation of:
 * Extended Bit Code Decimal Interchange Code
 * Extended Binary Coded Decimal Interchange Code ✓
 * Extended Bit Case Decimal Interchange Code
 * Extended Binary Case Decimal Interchange Code
- (iii) A bar-code reader is an example of a/an
 * Processing device
 * Storage device
 * Input device ✓
 * Output device
- (iv) This is the common language of Internet to develop a web page:
 * HTML ✓
 * BASIC
 * WWW
 * HTTP
- (v) This is the Fastest data communication medium:
 * Co-axial cable
 * Twister pair cable
 * Fiber optic cable ✓
 * Microwave
- (vi) In Full duplex transmission, data travels in:
 * parallel direction
 * two directions but not simultaneously
 * two directions but simultaneously ✓
 * one direction
- (vii) The internal process of starting up a computer is:
 * booting ✓
 * self-start
 * warm-up
 * start-up
- (viii) This register holds the instructions awaiting execution:
 * MCR
 * MBR
 * AC Register
 * Program Counter ✓
- (ix) A computer Virus is a:
 * bacteria
 * malware ✓
 * firmware
 * hardware
- (x) It controls all parts of the computer system:
 * ALU
 * CU ✓
 * MU
 * RAM
- (xi) The process of converting analog signals into digital signals is called:
 * Modulation
 * Demodulation ✓
 * Telecommunication
 * Communication
- (xii) Dos, Linux and Unix are the examples of:
 * operating system ✓
 * viruses
 * language translator
 * a program
- (xiii) Hardware and software combination used for connecting similar network is a:
 * router ✓
 * gateway
 * bridge
 * switch
- (xiv) A software that enables to display web pages is:
 * HTML
 * Web Browser ✓
 * www
 * Compiler
- (xv) This program translates whole program at a time:

- * Simulator
- * Compiler ✓
- * Assembler
- * Bus
- * Hybrid
- * Star
- * Laser ✓
- * Electrical
- * Secondary memory
- * Non-Volatile memory
- * External memory
- * Cable
- * Port
- * Bus
- * Ethernet
- * Magnetic Disk
- * Floppy Disk
- * DVD

- * bandwidth ✓
- * byte
- * frequency
- * connector
- * ports ✓
- * plugs
- * Internet ✓
- * Extranet
- * LAN
- * Character
- * Nibble ✓
- * Byte
- * Medium ✓
- * operating system
- * Protocol
- * Open System Interconnection ✓
- * Open System Internet
- * privacy
- * piracy ✓
- * photocopy
- * bug
- * vaccine
- * bomb

"B" (Short Answer Question) Marks: 30

pt any TEN part questions in all. All part questions
 rks. Answer should not exceed 6 to 7 lines
 nterpreter? How does it differ from a compiler?
 r
 ranslator which is used to convert programs in high-level
 evel language. Interpreter translates line by line and
 nce it encountered during the translation process.
FROM COMPILER
 Machine code is not stored anywhere, while in compiler
 e in the disk storage.

In interpreter The execution of the program takes place after every line is evaluated and hence the error is raised line by line if any, while in compiler The execution of the program happens only after the entire program is compiled.

Interpreted program runs slower., while Compiled program runs faster. The interpretation does not give any output program and is thus executed on every execution., while The compilation gives an output program which runs independently from the source file.

(ii) What is Data Processing Cycle? Write its steps with the diagram.

DATA PROCESSING CYCLE

Data processing cycle as the term suggests is sequence of steps or operations for processing data i.e. processing raw data to usable form. The processing of data can be done by number of data processing methods.

INPUT

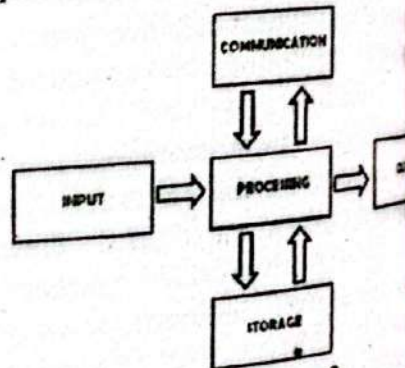
The raw data after collection needs to be fed in the cycle for processing. This is considered the first step and called input.

PROCESSING

Once the input is provided the raw data is processed by a suitable selected processing method. This is the most important step as it produces the processed data in form of output which will be used further.

OUTPUT

This is the final outcome and the raw data provided in the first stage is now "processed" and the data is useful and provides information and no longer called data.



(iii) What is the difference between Hardcopy and Softcopy?

ANSWER:

HARDCOPY	SOFTCOPY
Hard copies are present physically, and people can make use of them at different places	Soft copies do not exist physically.
Examples of hard copy are printed books, magazines, newspapers, comics, and other documents.	Examples of hard copy are PDF file, Office file and other programs file,
Hard copies do not depend on computers to be created or read.	Soft copies are dependent on computers.
Hard copies consume space and	Soft copies do not take any real

be carried at every place	space.
copies take a long time to be sent from one country to another	Soft copies can be sent anywhere in the world within few seconds.
copies can be converted into a paper with the help of scanner	Soft copy can be transformed into a paper with a photocopier.
copies are more expensive	Soft copies are cheaper.
copies have less additional costs.	Soft copies incur more extra costs.

ROM is called non-volatile memory?

ER: (See Question No. 2(vi), 2023)

say four abuses of Internet.

BULLYING:

using the internet to bully or intimidate, harass or threaten

NET FRAUD:

involves using the internet for the purpose of tricking victims out of money, properties and inheritance. Internet Fraud is popularly called "yahoo".

SPAM MAILS:

another way the internet can be abused and misused whereby bulk without verifiable link are sent to people in order to annoy them.

PLAGIARISM:

defined as using someone's ideas, words, music or video and passing it into yours, making it look like you are the original owner.

PIRACY:

one of the major abuse and misuse of the internet whereby someone uses or reproduces another person's work without being authorized.

Copyright Law. Give its advantages.

COPYRIGHT LAW

It is a legal term describing ownership of control of the rights to create and distribution of certain works of creative expression, including video, movies, music and computer programs. Historically, the law has been enacted to balance the desire of cultures to use and create works (thus creating "derivative work") against the desire of creators of art, literature, music and the like to monetize their work by controlling who can make and sell copies of the work.

PURPOSE OF COPYRIGHT LAW:

to create a fair and competitive environment.

2. Give networks operators the freedom to built their own back local access.
3. Facilitate rapid deployment of infrastructure for promotion services.
4. To enable a free society to function, minimum amount of instruction be permitted in terms of monitoring and filtering on all communication.

(vii) Define the main components of CPU.

CPU itself has following three components.

1. Memory or Storage Unit
2. Control Unit
3. ALU(Arithmetic Logic Unit)

1. MEMORY OR STORAGE UNIT

This unit can store instructions, data, and intermediate results. It supplies information to other units of the computer when needed. It is known as internal storage unit or the main memory or the primary or Random Access Memory (RAM).

2. CONTROL UNIT

This unit controls the operations of all parts of the computer but does not carry out any actual data processing operations. Its primary function is to carry out function such as, It is responsible for controlling the transfer of data and instructions among other units of a computer. It manages and coordinates all the units of the computer.

3. ALU(ARITHMETIC LOGIC UNIT)

This unit consists of following two subsections,

ARITHMETIC SECTION

Function of arithmetic section is to perform arithmetic operations such as addition, subtraction, multiplication, and division. All complex operations are done by making repetitive use of the above operations.

LOGIC SECTION

Function of logic section is to perform logic operations such as comparing, selecting, matching, and merging of data.

(viii) Draw OSI model.

ANSWER: (See Question No. 5, 2023)

(ix) Differentiate between asynchronous transmission and synchronous transmission.

ANSWER:

	ASYNCHRONOUS	SYNCHRONOUS
Form of Signal required for Transmission	In one form of asynchronous transmission, there is only one wire/signal	In synchronous transmission the stream of data to be transferred is encoded as fluctuating voltages in one

	carrying the transmission.	wire, and a periodic pulse of voltage is put on another wire (often called the "clock" or "strobe").
Transformation of Message Methods	sender provides a synchronization signal to the receiver before starting the transfer of each message	sender and receiver use the same clock signal
Properties	(i) Slower data transfer rate (ii) Does not need clock signal between the sender and the receiver	(i) supports high data transfer rate (ii) needs clock signal between the sender and the receiver
Communication Examples	This communication works much like the postal system	This communication works much like a phone call.

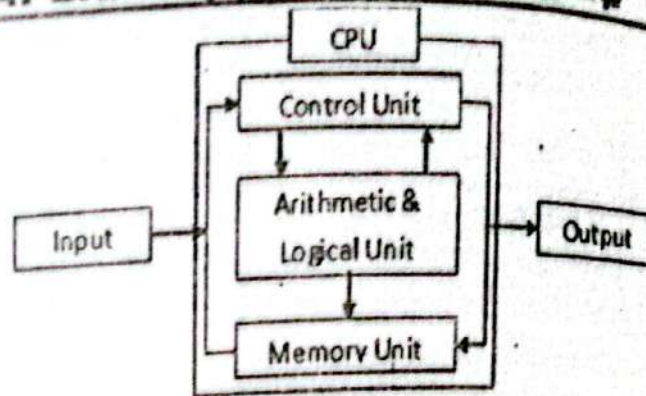
How computers can be useful in medical science?

ANSWER: Right from efficient data storage and easy access and sharing information, to conducting medical tests and simulating complex surgical procedures, computers have an important role to play in the medical field. Some important uses are following:

- Computer software is used for diagnosis of diseases. It can be used for examination of internal organs of the body. Advanced computer-based systems are used to examine organs of the body.
- The different types of monitoring equipment in hospitals are often based on computer programming.
- The functioning of hospital-bed beeping systems, emergency alarm systems, X-ray machines and several such medical appliances is based on computer logic.
- Computers are an excellent means for storage of patient-related data. Hospitals employ computer systems to maintain patient records.

Draw a label diagram of a digital computer.

ANSWER:

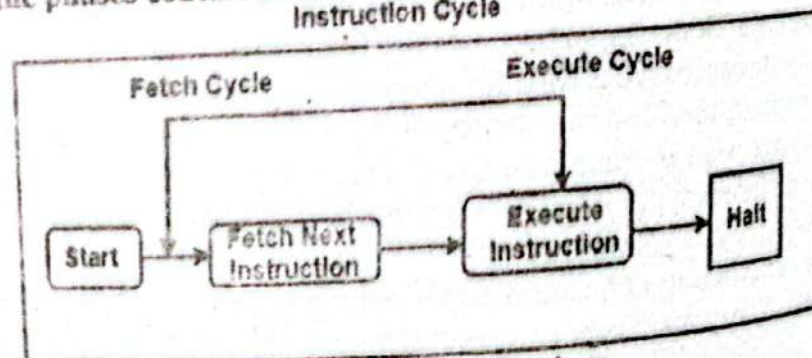


(xii) Define the instruction cycle with labeled diagram.

ANSWER: A program consisting of the memory unit of the computer includes a series of instructions. The program is implemented on the computer by going through a cycle for each instruction. In the basic computer, each instruction cycle includes the following procedures.

1. It can fetch instruction from memory.
2. It is used to decode the instruction.
3. It can read the effective address from memory if the instruction has an indirect address.
4. It can execute the instruction.

After the following four procedures are done, the control switches back to the first step and repeats the similar process for the next instruction. Therefore, the cycle continues until a Halt condition is met. The figure shows the phases contained in the instruction cycle.



(xiii) Write full form of any three of the following:

* MBR * ISO * HTTP * LED * OMR

ANSWER:

ABBREVIATION	FULL FORM
* MBR	Master Boot Record
* ISO	Open System Interconnection
* HTTP	Hypertext Transfer Protocol
* LED	Light-Emitting Diode
* OMR	Optical Mark Recognition

(xiv) Define the Computer Virus. Name any two viruses.

COMPUTER VIRUS

A virus is a program, script, or macro designed to cause damage, steal information, modify data, send e-mail, display messages, or perform any of these actions.

OF VIRUSES

Viruses: Star topology, and give its advantages.

TOPOLOGY

Star topology is designed with each node (like workstations, printers, servers etc.) connected directly to a central device called as a switch. Each workstation has a cable that goes from its network card (NIC) to a network switch.

ADVANTAGES OF STAR TOPOLOGY

Compared to Bus topology it gives far much better performance, as data necessarily get transmitted to all the workstations. A sent data reaches the intended destination after passing through no more than one and 2-3 links. Performance of the network is dependent on the central hub.

It is easy to connect new nodes or devices. In star topology new nodes can be added without affecting rest of the network. Similarly components can be removed easily.

It helps in monitoring the network. If one node or link doesn't affect the rest of network. At the same time it is easy to detect the failure and troubleshoot it.

Section C (Detailed - Answer Question) (16)

Attempt any TWO question from this section.

All questions carry equal marks.

Q.1. What is an Operating System? Explain the functions performed by an operating system.

ANSWER

An operating system or OS is a software program that enables the computer hardware to communicate and operate with the computer.

Without an operating system, a computer and software programs are useless. The operating system (OS) is the most important software that runs on a computer. Every general-purpose computer must have an operating system to run other programs and applications. Operating systems perform basic tasks, such as recognizing input from the keyboard, sending output to the display screen, keeping track of files and directories on the storage drives, and controlling peripheral devices, such as printers.

FUNCTIONS PERFORMED BY AN OPERATING SYSTEM
PROGRAMS EXECUTION

All operating systems can execute programs which mean they run programs that do something.

FILE MANAGEMENT

Computing devices that save data files have also file management capability. In Windows, this feature is called File Explorer.

CONNECTING TO INTERNET

Operating systems for computers support Internet connections.

I/O SYSTEM MANAGEMENT:

One of the main objects of any OS is to hide the peculiarities of hardware devices from the user.

SECONDARY-STORAGE MANAGEMENT:

Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be in primary storage or cache so that a running program can reference them.

4. Explain different types of communication media /channel used in communication.

TYPES OF COMMUNICATION MEDIA**1. TWISTED PAIR**

It consists of a pair of copper wires twisted around each other; they are around 1 to 2 mm thick and they are twisted to reduce the interference from the surrounding wires. Remember that a current carrying wire produces a magnetic field around it that can interfere with that of another wire in close proximity.

Invented by Alexander Graham Bell, this form of wire has been used since the 19th century and is currently the cheapest medium available. Twisted pairs are also used in telephone wires, but they only consist of two wires or two pairs. In computer networks, eight wires or four pairs are utilized. This is also known as the Ethernet cable or RJ-45 cable. The four pairs of wires are bundled together and covered by a protective shield.

2. COAXIAL CABLE

Offering better data rates and less signal attenuation, coaxial cables consist of a central copper conductor that is surrounded by a foil shield. The foil is covered by yet another shield known as a braided shield. Unlike twisted pairs, coaxial cables only have a single copper conductor. The central conductor and the foil shield are separated by a dielectric.

3. FIBER OPTIC

It is the most expensive of wired mediums and offers the highest data rates for long distance communication. They are often used in long distance communication and are never affected by any electromagnetic fields. This is because it involves light.

Optic cable is a thin, flexible, transparent medium made of very fine twisted pairs or coaxial cables, a fiber optic uses light pulses generated by laser or an injection diode to transmit data. Each pulse of light represents a single bit of data.

What is a Register? Explain the functions of various types of registers.

REGISTERS
Register (processor register OR CPU register) is one of a small set of data holding places that are part of the computer processor. Register are used to accept, store, and transfer data and instructions that are being processed immediately by the CPU, there are various types of Registers those used for various purpose.

Registers are the top of the memory hierarchy, and are the fastest for the system to manipulate data. In a very simple microprocessor, it consists of a single memory location, usually called an accumulator. Registers are built from fast multi-ported memory cell. They must be able to receive its data onto an internal bus in a single clock cycle. The result of an operation is stored here and could be re-used in a subsequent operation or saved into memory.

FUNCTIONS OF VARIOUS TYPES OF REGISTERS

MEMORY ADDRESS REGISTER (MAR):
This register holds the address of memory where CPU wants to read or write data. When CPU wants to store some data in the memory or reads data from the memory, it places the address of the required memory location in the MAR.

MEMORY BUFFER REGISTER (MBR):
This register holds the contents of data or instruction read from, or written to memory. The contents of instruction placed in this register are transferred to the Instruction Register, while the contents of data are transferred to the accumulator or I/O register.

I/O ADDRESS REGISTER (I/O AR):
Address register is used to specify the address of a particular I/O device.

I/O BUFFER REGISTER (I/O I3R):
Buffer Register is used for exchanging data between the I/O module and the processor.

PROGRAM COUNTER (PC)
Program Counter register is also known as Instruction Pointer Register. This register is used to store the address of the next instruction to be executed. When the instruction is fetched, the value of IP is incremented. Thus this register always points or holds the address of next instruction to be fetched.

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Once an instruction is fetched from main memory, it is stored in the Instruction Register. The control unit takes instruction from this register, decodes and executes it by sending signals to the appropriate component of computer to carry out the task.

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A stack represents a set of memory blocks; the data is stored in and retrieved from these blocks in an order, i.e. First In and Last Out (FILO). The Stack Control Register is used to manage the stacks in memory. The size of this register is 2 or 4 bytes.

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The Flag register is used to indicate occurrence of a certain condition during an operation of the CPU. It is a special purpose register with size one byte or two bytes. Each bit of the flag register constitutes a flag (or alarm), such that the bit value indicates if a specified condition was encountered while executing an instruction.

COMPUTER 2021

(100 Marks)

TIME: 2 Hours

NOTE:

- i) This section consists of 36 part questions and all are to be answered. Each question carries ONE marks.
- ii) Do not copy the part questions in your answer book. Write only the answer in full against the proper number of the question

SECTION 'A' (Multiple Choice Questions) (36)

1. Select the correct answer for each from the given options:

(i) This is the brain of computer:

- * ALU
- * Memory
- * CPU✓
- * Control unit

(ii) Integrated circuits (ICs) are related to this generation:

- * First generation
- * Second generation✓
- * Third generation✓
- * Fourth generation

(iii) CD-ROM is a:

- * Semiconductor memory
- * Memory register
- * Magnetic memor
- * Optical memory✓

(iv) A hybrid computer is :

digital computer
both a digital and an analogue computer✓

* analog computer

* tablet

This application program is used for creating presentation:

* MS Word

* MS Power Point✓

MS Access

MS Excel

This software resides permanently in a memory chip:

* Shareware

* Firmware✓

* Spyware

Freeware

Main component of second generation was:

* Transistor✓

* Diode

Integrated circuit

Micro chip

Number of logical OR & AND gates required for this logical equation

* 2(OR), 1(AND)✓

* 2(OR), 2(AND)

(A+B)(C+D) are:

1(OR), 1(AND)

1(OR), 2(AND)

The peripheral devices are attached with computer through:

* Connectors

* Plugs

* Sockets

Ports✓
This virus replicates itself to fill up memory space:

* TROJAN

* BOOT SECTOR VIRUS

WORM✓

TIME BOMB

It is the virus that activates on a particular date and time:

* Worm

* Boot sector virus

Time Bomb✓

Trojan

Conversion of Analog signal into digital signal is called:

* Demodulation

* Hashing

Modulation✓

Encryption

It is the logical gate that produces 1 output when all its inputs are set

1:

AND✓

* OR

* XOR

* NOT

The rule $A + AB$ is equal to:

* A✓

* A+B

* A.B

This is the most powerful computer system:

* Micro

* Super

* Analog

Mini

Artificial Intelligence is the key feature of:

Fourth Generation Computers

Second Generation Computers

Fifth Generation Computer✓

Third Generation Computers

It controls all parts of computer system:

* CU✓

* MU

* RAM

A printer which prints by striking paper is called:

* Laser printer.

* Impact Printer✓

Thermal printer

Non-impact printer

First generation computer used:

- * Transistor
- * Vacuum Tube ✓
- (xx) Any website can be viewed through:
 - * MS WORD
 - * Browser✓
- (xxi) A device to convert Analog signal into digital and vice versa is called:
 - * Diode
 - * Microprocessor
- (xxii) Full form of CD ROM is:
 - * MS POWER POINT
 - * Windows 10
- * Modern✓ * Compiler
- * Router * Bridge
- * Compact Disk Read Only Memory
- * Compact Data Read Only Memory
- * Compactable Disk Read Only Memory✓
- * Condense Disk Read Only Memory
- (xxiii) Printing equality of a printer is measured in:
 - * DPI✓ * Hertz * Mbps * Tbps
- (xxiv) A computer network designed in a single building is known as:
 - * LAN✓ * WAN * MAN * GAN
- (xxv) This program translates entire program at a time:
 - * Interpreter * Simulator * Compiler✓ * Assembler
- (xxvi) This logical gate needs a single 1 input to produce 1 output:
 - * OR✓ * AND * NOR * NOT
- (xxvii) RAM is also called:
 - * Secondary memory
 - * Main memory✓
 - * External memory
 - * Non-volatile
- (xxviii) Number of Bits is a nibble are:
 - * 2 * 4✓ * 16 * 32
- (xxix) In this topology all nodes are connected with a single cable:
 - * BUS✓ * STAR * MESH * TREE
- (xxx) It is not a kind of Display Monitor:
 - * LED * LCD * CRT * Plotter✓
- (xxxi) URL stands for:
 - * Unified Resource Location
 - * Universal Resource Locator
 - * Untraceable Resource Locator
 - * Uniform Resource Locator✓
- (xxxii) Processor speed is measured in:
 - * Mbps * Hertz ✓
- (xxxiii) This is not a logical gate:
 - * AND * OR * MOR✓ * NOT
- (xxxiv) Ink-jet is a kind of:
 - * Impact printer
 - * Thermal printer
 - * Non-impact printer✓
 - * Plotter
- (xxxv) This is the largest unit of memory:
 - * KB * MB * GB * TB✓
- (xxxvi) This is an example of Data Base management software:
 - * Java * MS-Access✓ * C-Language * Visual Basic

SECTION 'B' (Short-Answer Questions) (25)
 Answer any five part questions from this section. All
 carry equal marks. Answer should not exceed 6 to 7

distinction between Compiler and Interpreter

COMPILER	INTERPRETER
Line by line of the program is analyzed in	Line by line of the program is analyzed in an interpreter.
Machine code in the disk	Machine code is not stored anywhere.
Execution of the program takes place after the entire program is compiled.	The execution of the program takes place after every line is evaluated and hence the error is raised line by line if any.
Interpreted program runs faster	Interpreted program runs slower.
Interpretation gives an output program and is thus evaluated on every execution.	The interpretation does not give any output program and is thus evaluated on every execution.

any two laws of Boolean Algebra

BOOLEAN ALGEBRA

COMMUTATIVE LAW

This Law states that the interchanging of the order of operands in an equation does not change its result.

$$A + B = B + A$$

$$A * B = B * A$$

DISTRIBUTIVE LAW

This Law of multiplication states that the AND operation are done first before the OR operation when more than two variables.

$$A * (B + C) = (A * B) + (A * C)$$

non-volatile memory. Name any three.

NON-VOLATILE MEMORY

Non-volatile memory (NVM) is a type of memory that retains stored data even when power is turned off. Unlike volatile memory, it does not require an external power source to maintain the storage state. Only reading and writing data to non-volatile memory requires power.

NON-VOLATILE MEMORY

1. Hard disk drive (HDD)
2. Solid state drive (SSD)
3. Flash drive (USB keychain)
2. (iv) Write Full-Form of the following:
 * DIMM * CRT * BIOS * PROM

ANSWER:

ABBREVIATION	FULL FORMS
DIMM	Dual In-Line Memory Module
CRT	Cathode Ray Tube
BIOS	Basic Input / Output System
PROM	Programmable Read Only Memory

2. (v) Define the function of Control unit in CPU

FUNCTIONS

ANSWER:

The functions of the control unit include the following.

1. It directs the flow of data sequence between the processor and devices.
2. It can interpret the instructions and controls the flow of data in processor.
3. It generates the sequence of control signals from the received instructions or commands from the instruction register.
4. It has the responsibility to control the execution units such as ALU, buffers, and registers in the CPU of a computer.
5. It has the ability to fetch, decode, handle the execution, and store results.
6. It cannot process and store the data.
7. To transfer the data, it communicates with the input and output devices and controls all the units of the computer.

2. (vi) Why Operating System is the most important software in computer system.

IMPORTANCE OF OPERATING SYSTEM:

An operating system is the most important software that runs on a computer. It manages the computer's memory and processes, as well as of its software and hardware. It also allows you to communicate with a computer without knowing how to speak the computer's language.

2. (vii) Why do we need to keep antivirus updated?

ANSWER: It is important to constantly update the antivirus software on a computer because computers are regularly threatened by new viruses. The anti-virus updates contain the latest files needed to combat new viruses and protect your computer.

Antivirus software provides signature files which are very important since they contain the latest lists of known viruses. These signature files are released daily, and sometimes even more often.

Cache memory is different from RAM.

Cache Memory is Different from RAM

Cache is a component in the computer that stores data so that future data that data can be served faster while RAM is a form of data storage that stores data and machine code currently being used.

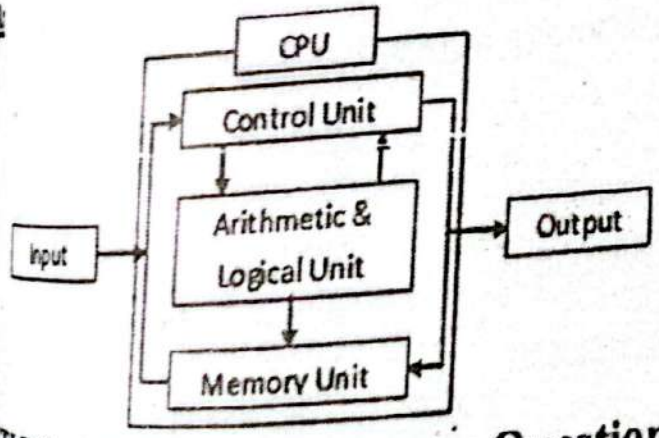
Cache is faster than secondary storage mediums, but it is not as fast as RAM.

Capacity of cache is smaller than RAM, which has a higher capacity. Cache holds frequently used data by the CPU while RAM holds data that are currently executed by the CPU.

Use of MODEM

Users need a modem to connect to the Internet because signals that travel through telephone and cable lines are analog, and computers can only process digital information. Modem converts information from digital to analog mode at the transmitting end and demodulator converts from analog to digital at receiving end. The process of converting signals of one computer network into digital signals of another network so they can be processed by a receiving computer is called digitizing.

Block diagram of Computer System.



SECTION "C" (Detailed Answer Questions)(14)

Attempt any Two questions from this section. All carry equal marks.

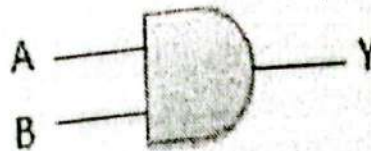
Answer logic gates with diagrams.

LOGIC GATES

A logic gate which performs an AND operation is shown in figure. It has n inputs and one output.

Y = A AND B AND C N
 Y = A.B.C N
 Y = ABC N

LOGIC DIAGRAM

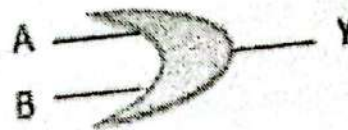


OR Gate

A circuit which performs an OR operation is shown in figure. It has n input ($n \geq 2$) and one output.

Y = A OR B OR C N
 Y = A + B + C N

LOGIC DIAGRAM

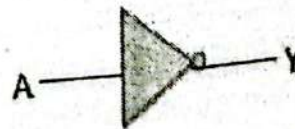


NOT Gate

NOT gate is also known as Inverter. It has one input A and one output Y

Y = NOT A
 Y = $\overline{A}$
 Y = NOT A
 Y = $\overline{A}$

LOGIC DIAGRAM

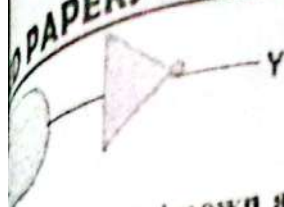


NAND Gate

A NOT-AND operation is known as NAND operation. It has n input ($n \geq 2$) and one output.

Y = A NOT AND B NOT AND C N
 Y = A NAND B NAND C N

LOGIC DIAGRAM

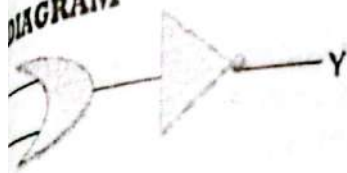


operation is known as NOR operation. It has n input ($n \geq 2$) and one output.

$$= A \text{ NOT OR } B \text{ NOT OR } C \dots\dots N$$

$$= A \text{ NOR } B \text{ NOR } C \dots\dots N$$

DIAGRAM



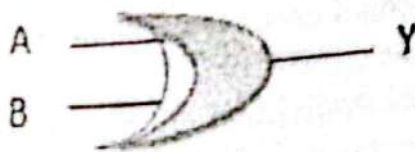
is-OR gate is a special type of gate. It can be used in the half adder and subtractor. The exclusive-OR gate is abbreviated as X-OR or sometime as X-OR gate. It has n input ($n \geq 2$) and one output.

$$= A \text{ XOR } B \text{ XOR } C \dots\dots N$$

$$= A \oplus B \oplus C \dots\dots N$$

$$= \overline{AB} + \overline{AB}$$

DIAGRAM



is a special type of gate. It can be used in the half adder, full adder, subtract or. The exclusive-NOR gate is abbreviated as EX-NOR or sometime as X-NOR gate. It has n input ($n \geq 2$) and one output.

$$= A \text{ XOR } B \text{ XOR } C \dots\dots N$$

$$= A \ominus B \ominus C \dots\dots N$$

$$= \overline{AB} + \overline{AB}$$

DIAGRAM



4. What is printer? Discuss different types of Impact printers.

ANSWER:

PRINTER

Printers are Output devices used to prepare permanent Output device paper.

TYPES OF IMPACT PRINTERS

ANSWER: (See Question No. 4, 2024)

5. What is Computer Virus? Explain its different types.

ANSWER: (See Question No. 7, 2024)

(iii) What are the functions of Router and Gateway in Data Communication?

ROUTER

A router (including a wireless router) is a specialized networking device connected to two or more networks running software that allows the router to move data from one network to another. Router functions in an Internet protocol based network operate at the network layer (OSI Model layer 3). The primary function of a router is to connect networks together and keep certain kinds of broadcast traffic under control.

GATEWAY

A gateway is designed to safeguard a private network, protecting it from malicious traffic from the outside world and granting or denying access to computers within the network to the wider world. A gateway and a router can be combined. A router's job is to forward packets of data from one computer to another throughout the network.

FUNCTIONS OF DATA COMMUNICATION:

The data transmission software or protocols perform the following functions for the efficient and error free transmission of data.

DATA SEQUENCING:

The function of a protocol to divide the long message into smaller packets of fixed size that are to be transmitted for error free data transmission is called Data Sequencing.

DATA ROUTING:

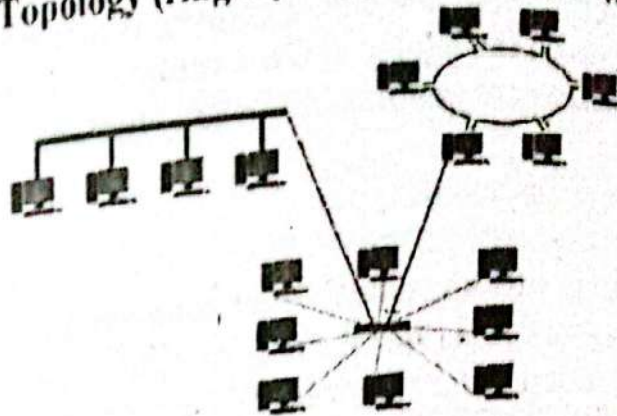
The function of a protocol to find the most efficient path or route between the sender and the receiver before sending the data is called Data Routing.

FLOW CONTROL:

The function of a protocol to control the rate of data transmission from the sender to the receiver is called Flow Control. It regulates the process of sending data between fast sender and slow receiver.

ERROR CONTROL:

used and in another star topology is used, connecting these topologies result in Hybrid Topology (ring topology and star topology).



سال 2020 میں گیارہویں اور بارہویں جماعت کے امتحانات نہیں ہوئے تھے۔

COMPUTER 2019

Max. Marks: 2

Time: 20 minutes

SECTION 'A' (MULTIPLE CHOICE QUESTION)

1. Choose the correct answer for each from the given options:

- (i) It is also known as Memory Data Register:
 - *Memory Address Register
 - *Instruction Register
 - *Memory Buffer Register ✓
 - *Program Counter
- (ii) FM stands for:
 - *Frequency Modification
 - *Frequency module
 - *Frequency Modulation ✓
 - *Functional module
- (iii) it is the most common network used to connect two or more computers which are lying in the same office or building:
 - *GAN
 - *MAN
 - *LAN ✓
 - *WAN
- (iv) The physical layout of a LAN is called:
 - *link
 - *layer cake
 - *contention
 - *topology ✓
- (v) This is the fastest data communication media:
 - *Coaxial Cable
 - *Twisted Pair
 - *Microwave
 - *Fibre Optic ✓
- (vi) The illegal copy of copyright software is called:
 - *Theft of Software
 - *Hacking
 - *Software Piracy ✓
 - *Computer crime
- (vii) Dot Matrix is a type of:
 - *Impact Printer ✓
 - *Non-impact Printer
 - *Scanner
 - *Plotter
- (viii) DOS, Linux and Unix are the types of:
 - *Operating Systems ✓
 - *Translators
 - *Viruses
 - *Language Translators
- (ix) This is a computer crime:
 - *Formatting
 - *Backup
 - *Restoring
 - *Hacking ✓
- (x) This software is stored permanently on ROM chip by manufacturer

- *Firmware
- *Hardware
- *Software
- *Simplex Mode
- *Half-simplex Mode
- *Full-duplex Mode
- *Application Software
- *Malware
- *Storage Unit
- *Processing Unit
- *byte
- *string
- *unit
- *Control bus
- *central bus
- *address bus

COMPUTER 2019

Marks: 60

2 Hours 40 Minutes

SECTION 'B' (SHORT-ANSWER QUESTIONS) (36)

Answer Twelve part questions from this section. All answers carry equal marks.

Write the name of the Protocol and name any three.

Q. 2 (xiv), 2023

Write any two of the following:

*Router

*Gateway

A bridge is a repeater, with the ability of filtering content by reading the MAC addresses of source and destination. It is also used for interconnecting two LANs working on the same protocol. It has a single input and single output port, thus making it a 2 port device.

A router is a device like a switch that routes data packets based on their IP addresses. Router is mainly a Network Layer device. Routers normally connect different WANs together and have a dynamically updating routing table in which they make decisions on routing the data packets. Router divides a network into domains of hosts connected through it.

A gateway, as the name suggests, is a passage to connect two networks together. It works upon different networking models. They basically work as the agents that take data from one system, interpret it, and transfer it to another system. Gateways are also called protocol converters and can work at any network layer. Gateways are generally more complex than routers.

(iii) Which mode of Data transmission is suitable for communication from a computer to a printer?

ANSWER: Simplex transmission mode is suitable for communication from a computer to a printer. In a simplex transmission mode, the communication between sender and receiver occurs only in one direction. The sender can only send the data and the receiver can only receive the data.

(iv) Write the full forms of the followings:

* HTTP

* URL

* SIMM

ABBREVIATION	FULL FORMS
HTTP	Hypertext Transfer Protocol
URL	Simple Inline Memory Module
SIMM	Simple Inline Memory Module

(v) Define Ring topology and give its advantages.

RING TOPOLOGY

A ring topology is a network configuration in which device connections create a circular data path. Each networked device is connected to two others, like points on a circle. Together, devices in a ring topology are referred to as a ring network.

ADVANTAGES OF RING TOPOLOGY

All data flows in one direction, reducing the chance of packet collisions. A network server is not needed to control network connectivity between each workstation.

Data can transfer between workstations at high speeds. Additional workstations can be added without impacting performance of the network.

(vi) Differentiate between Volatile memory & Non-Volatile memory.

ANSWER:

VOLATILE MEMORY	NON-VOLATILE MEMORY
> Volatile Memory is the type of computer memory that is temporary in nature. It stores the data inside it only until the power is supplied. > Volatile memory store data of the programs that are currently in process by the CPU. Frequently used data and information about the process is stored in Volatile memory. > Volatile memories are the fastest memories in nature. They contain frequently used	> Non-Volatile Memory is the type of computer memory that is permanent in nature. The data stored in such kind of memory remains there even after the system is turned off. > Non-volatile memory store data of the basic booting process of the computer system BIOS. All kind of media and data that has to be saved permanently or for the longer period. > Non-Volatile memories whereas are relatively slow as compare to volatile memories.

inside them and data is
processed from them quickest

The data accessed from non-
volatile memory is slow as
compared to volatile memory.

Impact Printer and Non-Impact Printer.

PRINTER

Impact printer has mechanisms resembling those of a typewriter. It forms
characters or images by striking a mechanism such as a print hammer or
stylus against an inked ribbon, leaving an image on paper.

NON-IMPACT PRINTERS

Non-impact printers, used almost everywhere now, are faster and quieter than
impact printers because they have fewer moving parts. Nonimpact printers
form characters and images without direct physical contact between the
printing mechanism and the paper.

Analog Single and Digital Signal.

SINGLE

A single signal is any continuous signal for which the time varying feature
of the signal is a representation of some other time varying quantity,
as opposed to another time varying signal. It differs from a digital signal in
that it has small fluctuations in the signal which are meaningful.

DIGITAL SIGNAL

A digital signal uses discrete (discontinuous) values. By contrast, non-digital
signals use a continuous range of values to represent information.
If digital representations are discrete, the information represented can
be discrete, such as numbers or letters, or continuous, such as sounds,
or other measurements of continuous systems.

Block diagram of a Digital Computer.

(See Question No. 2 (xi), 2022)

Is Internet useful in the field of education?

Internet hinders students from doing their homework or assigned
work on their own. Thanks to internet, students can obtain the information they
need with great ease by simply clicking the button. However, this great
ease to any kind of information can be abused. For instance, when
given an assignment, what many of them do is just go on the
internet, find the blogs where the seemingly needed information is posted, and
copy and paste it without looking the material thoroughly later. Some
even just download the material which someone else has already done
for them as if it were their work.

What is an Address Buss called unidirectional?

A Data bus is bidirectional because it transfers information between
devices and memory register but; address bus is unidirectional
because it only receives information.

What is an Operating System necessary for a computer?

Without a computer operating system, a computer and software
would be useless. The operating system (OS) is the most important

program that runs on a computer. Every general-purpose computer must have an operating system to run other programs and applications. Computer operating systems perform basic tasks, such as recognizing input from a keyboard, sending output to the display screen, keeping track of files and directories on the storage drives, and controlling peripheral devices, such as printers.

(xiii) What are the components of LAN?

COMPONENTS OF LAN:

Components of LAN are following:

- Network Cards
- Network Cables
- Network Hubs
- Network Switches
- Bridge
- Routers
- Gateway

(xiv) What are the basic hardware needed for Internet?

BASIC HARDWARE FOR INTERNET

The basic computer hardware components that are needed to set up a network are as follows:

1. Network Cables
2. Routers
3. Repeaters, Hubs, and Switches
4. Bridges
5. Gateways
6. Network Interface Cards

(xv) Name any two Full-duplex devices.

FULL-DUPLEX DEVICES

- Telephone or Mobile Phone
- Router

(xvi) What is Computer crime?

COMPUTER CRIME

Computer crime is an act performed by a knowledgeable computer user, sometimes referred to as a hacker that illegally browses or steals a company or individuals private information. In some cases, this person or group of individuals may be malicious and destroy or otherwise corrupt the computer or data files.

(xvii) Draw the layer cake of OSI model.

ANSWER: (See Question No. 2 (viii) 2022)

(xviii) Match the terms of column A with those of column B:

Column "A"	Column "B"
Keyboard	Operating System
Non-Volatile	Firmware
Byte	LAN
DOS	ROM
BIOS	8 bits
Local Area Network	Input device

ANSWER:

555 COMPUTER

Column "A"	Column "B"
Keyboard	Input device
Non-Volatile	ROM
Byte	8 bits
DOS	Operating System
BIOS	Firmware
Local Area Network	LAN

QNC'(DETAILED-ANSWER QUESTIONS)(24)

Answer 2 Questions from this section. All questions carry marks.

What is Internet Browser? Describe its various types.

NET BROWSER

Net Browser is an application program that provides a way to look at and interact with all the information on the World Wide Web. The word "browser" seems to have originated prior to the Web as a generic term for interfaces that let our browse (navigate through and read) text files.

Basically, a Web browser is a client program that uses HTTP (Hypertext Transfer Protocol) to make requests of Web servers throughout the Internet. If of the browser user. Most browsers support e-mail and the File Transfer Protocol (FTP) but a Web browser is not required for those Internet services and more specialized client programs are more popular.

100% INTERNET BROWSER

INTERNET EXPLORER

Internet Explorer (IE) is a product from software giant Microsoft. This is the most commonly used browser in the universe. This was introduced in 1995 with Windows 95 launch and it has passed Netscape's popularity in 1998.

GOOGLE CHROME

This browser is developed by Google and its beta version was first released on September 2, 2008 for Microsoft Windows. Today, chrome is expected to be one of the most popular web browser with its global share of more than 15%.

APPLE SAFARI

Safari is a new browser derived from Mozilla. It was released in 2004 and has become the second most popular browser on the Internet.

INTERNET BROWSERS SAFARI

Safari is a web browser developed by Apple Inc. and included in Mac OS X. It was released as a public beta in January 2003. Safari has very good support for latest technologies like XHTML, CSS2 etc.

INTERNET BROWSER OPERA

Opera is smaller and faster than most other browsers, yet it is full-featured. It is user-friendly, with keyboard interface, multiple windows, zoom

functions, and more, Java and non Java-enabled versions available. Ideal for newcomers to the Internet, school children, handicap and as a front-end for CD-Rom and kiosks.

3. (b) What is a Computer Virus? What preventive measures can be taken to protect our computer from its attacks?

COMPUTER VIRUS

A computer virus is a program, script, or macro designed to cause damage, steal personal information, modify data, send e-mail, display messages, or some combination of these actions.

When the virus is executed, it spreads by copying itself into or over data files, programs, or boot sector of a computer's hard drive, or potentially anything else writable. To help spread an infection the virus writers use detailed knowledge of security vulnerabilities, zero days, or social engineering to gain access to a host's computer.

PROTECTION OF COMPUTER:

The only solution to never risk getting a computer virus is to keep the computer disconnected from the internet and off which is not use.

We have many options that can greatly reduce the risk of getting a computer virus.

- Keep our operating system up to date. Microsoft, Apple and Linux have updates that roll out on a regular basis. Keeping our operating system up to date alone will stop the vast majority of known threats.
- Keep our web browser updated. Nearly all web browsers have an auto update feature that allows us to auto update our web browser. Popular browsers out there include, FireFox, Chrome, Safari and Edge (formerly known as Internet Explorer)
- Keep our software programs updated. Programs such as Office, Adobe PDF, Flash are all targeted programs due to their wide user base. The more popular a program is the more it's a target for exploits.
- Safe Web Browsing. This is covered more in-depth in the Safe Web Browsing section. Basically don't un-trusted websites and we will be 99% of the way there.
- Have an Active antivirus client installed that prevents virus infections and keep the program updated.
- Do not open e-mails with attachments from unknown senders. Even if it's a known use caution as their e-mail could of been hacked.

3. (c) Define the term Bus. Describe the functions of any two Buses.

BUS

A bus is a common pathway through which information flows from one computer component to another. This pathway is used for communication purpose and it is established between two or more computer components.

FUNCTIONS

DATA BUS:

- The data bus allows data to be transferred between the memory and the processor.

The data bus is bi-directional as data must go both ways.
 The data bus is made up of a set of wires or lines; each line takes one bit of the data.
 The width of the data bus usually matches the size of a memory location, processor registers are also usually the same size.
 The width of the data bus is measured by the number of lines on the bus.
 A 32-bit data bus has 32 lines and can transfer 32-bit of data in one go.
 The width of the data bus is a factor in system performance.
 The wider the data bus the more data that can be transferred in one go.

ADDRESS BUS

A memory location is identified by its unique address.
 The address bus holds the address of the memory location being accessed.
 The address bus is made up of a set of wires or lines; each line takes one bit of the address.
 The address bus is uni-directional as it is used by the processor to inform memory which addresses it needs. (Memory can't "send" an address to processor!)
 The width of the address bus is measured by the number of lines on the bus.
 A 32-bit address bus has 32 lines and can identify 2^{32} memory locations.
 The wider the address bus the more memory locations can be identified.

(a) State Copyright law. Give its advantages.

ANSWER: (See Question No. 2 (vi), 2022)

(b) What is Stack? Explain its operations with examples.

STACK

Stack is a concept mainly used in data structures. Stack is used with the array concept and used for data inserting, modifying and recovering purpose. Stack use the concept of LIFO (Last-In-First-Out). LIFO refers, as the stack operations will be performed on the data that is in the last segment of the array. Stack can also be used in linked-list but at the minimum level compared with array.

OPERATIONS OF STACK

Stack generally performs three operations out of which the first two are the important ones. The operations of the stack are as follows:

1. Push

2. Pop

PUSH

Push operation is used to insert an item in the stack. Push operation also performs whether the stack is empty and if it is empty then Push operation will add or insert an item in to the stack. Once the stack is full and there is no memory space to add an item, the push operation will inform that the stack is in overflow state. So, no new item can be inserted into the stack.

POP

Pop operation is used to delete or remove an item from the stack. Pop operation also performs whether the stack is full or not. If the stack is full, the pop operation will delete the item one by one from the stack till the stack reach

empty. Once the stack is empty, Pop operation will return the message as the stack is in underflow state. So, there is no item in the stack to delete or remove.

4. (c) Define System Software and explain its three categories.

SYSTEM SOFTWARE

System software is software on a computer that is designed to control and work with computer hardware. The two main types of system software are the operating system and the software installed with the operating system, often called utility software. The operating system and utility software typically depend on each other to function properly.

TYPES OF SYSTEM SOFTWARE

The System Software are divided into 3 types, They are :

1. Operating System Software
2. Language Translator
3. Utility Software

1. OPERATING SYSTEM SOFTWARE (O.S) :

An Operating System is a program designed to run other programs on a computer. A computer's operating system is its most important program. It is considered as the backbone of a computer managing both software and hardware device. Operating systems are responsible for each and everything from the control and allocation of memory to input from external devices and output to computer display.

An operating system also plays a vital role in security. Its job includes preventing unauthorized user from accessing the computer system.

There various types of operating system are -

- | | |
|------------------------|-------------------------|
| ➤ General Purpose O.S | ➤ Multi-Programming O.S |
| ➤ Special Purpose O.S | ➤ Multi-Processing O.S |
| ➤ Batch Processing O.S | ➤ Multi-Threading O.S |
| ➤ Single Use O.S | ➤ Real Time O.S |
| ➤ Multi-User O.S | ➤ Embedded O.S |

2. LANGUAGE TRANSLATOR :

It is another system software which convert the high level language to machine level language for the purpose of machine understanding.

The Machine can only understand the machine level language or binary language 0's & 1's. The language translator rectify the errors within the program through different ways.

There are 3 types of language translator, they are as follows :-

- Compiler.
- Interpreter.
- Assembler.

3. UTILITY SOFTWARE :

Utility Software is a kind of system software designed to help, analyse, configure, optimize and maintain the computer. A single piece of utility software is usually called a utility or tool.

If we are having any problem related to system then we can solve it through utility software.

Antivirus:- Utility scan for computer virus. Memory tester
memory failure etc.
are the activities those are coming under the utility software are

Up Recovery.
Defragmenter.
Cleaners.
Partitions. Etc

SYNCHRONOUS & ASYNCHRONOUS TRANSMISSIONS WITH DIAGRAMS.

SYNCHRONOUS TRANSMISSION
In synchronous transmission, data moves in a complete paired approach in the chunks or frames. Synchronisation between the source and target is so that the source knows where the new byte begins since there is no gap between the data.
Synchronous transmission is effective, dependable and is utilised for sending a large amount of data. It offers real-time communication between linked devices.

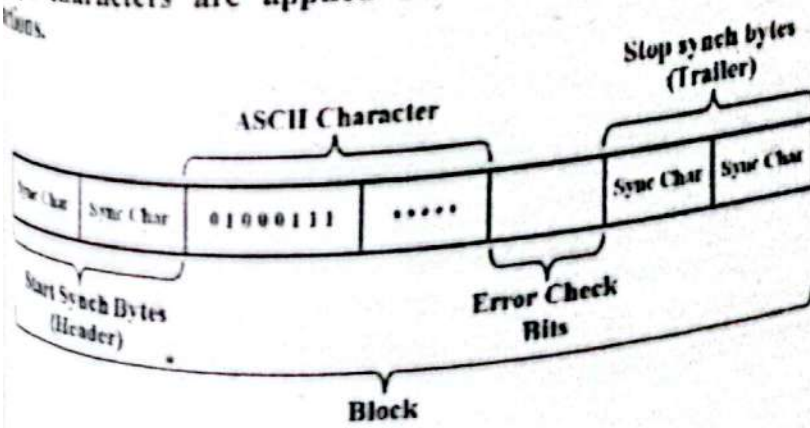
In case of synchronous transmission would be the transfer of a large file. Before the file is transmitted, it is first dissected into blocks of data. The blocks are then transferred over the communication link to the destination.

Since there is no beginning and end bits the data transfer rate is quicker but there is a possibility of more errors to occur. Over time, clocks will get out of sync at the target device would have the incorrect time, so some bytes could be corrupted due to lost bits. To resolve this issue, there is a need for re-synchronisation of the clocks as well as the use of check digits to ensure that the bytes are correctly received and translated.

CHARACTERISTICS OF SYNCHRONOUS TRANSMISSION

There are no spaces in between characters being sent. Timing is provided by modems or other devices at the end of the transmission.

Each syn character goes before the data being sent. The syn characters are applied between chunks of data for timing purposes.



ASYNCHRONOUS TRANSMISSION

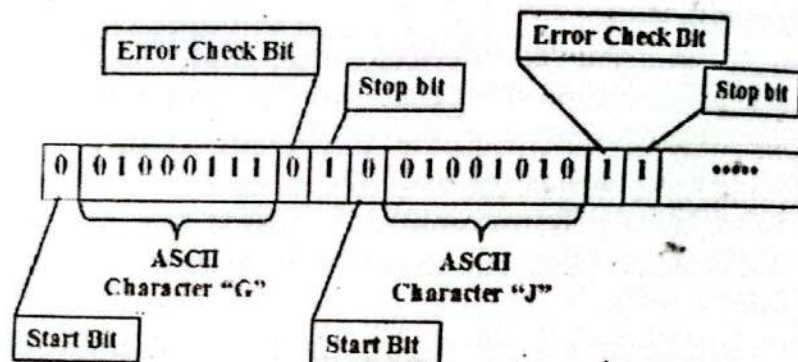
In asynchronous transmission data moves in a half-paired approach, 1 byte of a character transmitted is 8 bits where a parity bit is added each at the beginning and at the end which makes it a total of 10 bits. It doesn't need a clock for integration; rather it utilises the parity bits to inform the receiver how to translate the data.

It is straightforward, quick, cost-effective and doesn't need a 2-way communication.

CHARACTERISTICS OF ASYNCHRONOUS TRANSMISSION

- Each character is headed by a beginning bit and superseded by one or more end bits.
- There may be gaps or spaces in between characters.

ASYNCHRONOUS TRANSMISSION



5. (b) Explain the important features of an Operating System.

FEATURES OF OPERATING SYSTEM

Here is a list commonly found important features of an Operating System:

PROGRAMS EXECUTION

All operating systems can execute programs which mean they can run programs or applications. They run programs that do something useful.

FILE MANAGEMENT

Computing devices that save data files have also file management capability. In Windows, this feature is called File Explorer.

CONNECTING TO INTERNET

Operating systems for computers support Internet connections.

I/O SYSTEM MANAGEMENT:

One of the main objects of any OS is to hide the peculiarities of that hardware devices from the user.

SECONDARY-STORAGE MANAGEMENT:

Systems have several levels of storage which includes primary storage, secondary storage, and cache storage. Instructions and data must be stored in primary storage or cache so that a running program can reference it.

5. (c) What is meant by Primary and Secondary Storages? Define any three types of secondary storage.

PRIMARY STORAGE

Primary storage, also known as main storage or memory, is the main area in a computer in which data is stored for quick access by the computer's processor. Information must be transferred to primary storage. On today's smaller computers, especially personal computers and workstations, the term random access memory (RAM) - or just memory - is used instead of primary, main or core memory, or immediate access storage, internal storage, and the like. Diskette, CD, and DVD collectively describe secondary storage or secondary storage.

SECONDARY STORAGE

Secondary memory is also known as secondary storage. The secondary memory is accessed indirectly via input/output operations. This memory is also permanent, external, stable or persistent memory. It is characterised by its access and cheapness, relative to the RAM, and by its permanent storage.

TYPES OF SECONDARY STORAGE**1. DISK**

A hard disk is a form of auxiliary storage. A hard disk consists of more rigid metal plates coated with a metal oxide material that allows data to be magnetically recorded on the surface of the platters. The hard disk spins at a high rate of speed, typically 5400 to 7200 revolutions per minute (RPM). Storage capacities of hard disks for personal computers range from 1 GB to 120 GB (one billion bytes are called a gigabyte).

2. OPTICAL DISKS

Mass Storage Devices Store bit values as variations in light reflection. They have higher area density & longer data life than magnetic storage. They are standardized and relatively inexpensive. Their Uses: read-only storage for performance requirements, applications with high capacity requirements & where portability in a standardized format is needed.

1. Optical Disk**1. ROM (read only)**

2. CD-R: (record) to a CD.

CD-RW: can write and erase CD to reuse it (re-writable)

2. Digital Video Disk**3. FLOPPY DISK**

Floppy disks are small removable disks that are plastic coated with magnetic recording media. Floppy disks are typically 3.5" in size (diameter) and can hold 1.44 MB of data. This portable storage device is a rewritable media and can be used a number of times. Floppy disks are commonly used to move files between different computers. The main disadvantage of floppy disks is that they are damaged easily and, therefore, are not very reliable. The following figure is an example of the floppy disk. Figure 3 shows a picture of the floppy disk.